

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 4

8 Semantics of Programming Languages (nk480)

Consider the following grammar of types and subtyping relation:

$$A, B ::= \text{int} \mid \{l_1 : A_1, \dots, l_n : A_n\} \mid A \rightarrow B$$

$$\frac{}{\text{int} \leq \text{int}} \leq_{\text{int}}$$

$$\frac{B_1 \leq A_1 \quad A_2 \leq B_2}{A_1 \rightarrow A_2 \leq B_1 \rightarrow B_2} \leq_{\rightarrow}$$

$$\frac{n \leq m \quad A_1 \leq B_1 \quad \dots \quad A_n \leq B_n}{\{l_1 : A_1, \dots, l_n : A_n\} \leq \{l_1 : B_1, \dots, l_m : B_m\}} \leq_{\text{rec}}$$

We did not include explicit rules for reflexivity or transitivity in our subtyping relation. However, we can prove that they are *admissible* – that is, the subtyping judgement already has enough power to simulate them.

In the following questions, you may make use of any needed inversion properties without giving explicit proofs for them.

- (a) Prove that for each type A , a subtyping derivation $A \leq A$ can be constructed. [5 marks]

Answer: This proof follows by induction on A .

- Case $A = \text{int}$.

In this case, the result follows by rule $\leq_{\text{int}}$

- Case $A = A_1 \rightarrow A_2$.

(i) By induction, we know that $A_1 \leq A_1$.

(ii) By induction, we know that $A_2 \leq A_2$.

(iii) Therefore, by rule $\leq_{\rightarrow}$ on (i) and (ii), we derive $A_1 \rightarrow A_2 \leq A_1 \rightarrow A_2$.

- Case $A = \{l_1 : A_1, \dots, l_n : A_n\}$.

(i) For each $i \leq n$, we know $A_i \leq A_i$ by induction on A_i .

(ii) Therefore, by rule $\leq_{\text{rec}}$, we can derive $\{l_1 : A_1, \dots, l_n : A_n\} \leq \{l_1 : A_1, \dots, l_n : A_n\}$

- (b) Prove that for all A, B, C , if $A \leq B$ and $B \leq C$ are derivable, then $A \leq C$ is derivable. [15 marks]

Answer: There are multiple ways of proving this, each of which exploit slightly different inversion properties. I'll choose induction on B .

- Case $B = \text{int}$.

- (i) By inversion on the derivation of $A \leq \text{int}$, we know $A = \text{int}$.
 - (ii) By inversion on the derivation of $\text{int} \leq C$, we know $C = \text{int}$.
 - (iii) By rule $\leq \text{int}$, we know $\text{int} \leq \text{int}$, which shows $A \leq C$.
 - Case $B = B_1 \rightarrow B_2$:
 - (i) By inversion on the derivation of $A \leq B_1 \rightarrow B_2$, we know that $A = A_1 \rightarrow A_2$ and $B_1 \leq A_1$ and $A_2 \leq B_2$.
 - (ii) By inversion on the derivation of $B_1 \rightarrow B_2 \leq C$, we know that $C = C_1 \rightarrow C_2$ and $C_1 \leq B_1$ and $B_2 \leq C_2$.
 - (iii) By induction on $C_1 \leq B_1$ and $B_1 \leq A_1$, we get $C_1 \leq A_1$.
 - (iv) By induction on $A_2 \leq B_2$ and $B_2 \leq C_2$, we get $A_2 \leq C_2$.
 - (v) Hence by rule $\leq \rightarrow$, we get $A_1 \rightarrow A_2 \leq C_1 \rightarrow C_2$.
 - (vi) Hence we know $A \leq C$.
 - $B = \{l_1 : B_1, \dots, l_n : B_n\}$
 - (i) By inversion on $A \leq B$, we know $A = \{l_1 : A_1, \dots, l_m : A_m\}$, and $m \leq n$ and $A_i \leq B_i$ for each $i \leq m$.
 - (ii) By inversion on $B \leq C$, we know $C = \{l_1 : C_1, \dots, l_k : C_k\}$, and $n \leq k$ and $B_i \leq C_i$ for each $i \leq k$.
 - (iii) Since $m \leq n \leq k$, we know that $A_i \leq B_i$ and $B_i \leq C_i$ for each $i \leq m$.
 - (iv) For each $i \leq k$, we know that $A_i \leq C_i$ by induction on $A_i \leq B_i$ and $B_i \leq C_i$.
 - (v) Since $m \leq k$ and $A_i \leq C_i$ for each $i \leq k$, by rule $\leq \text{rec}$ we get $\{l_1 : A_1, \dots, l_m : A_m\} \leq \{l_1 : C_1, \dots, l_k : C_k\}$.
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